SPbUnited TDP

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***Abstract*—The report provides an overview of the robot control system of the St. Petersburg SPb United team within the RoboCup SSL robotics competition. The control system implements a hierarchical approach, dividing management tasks into multiple levels, each responsible for specific functions. Key levels include Strategy, Router, high-level (RobotHI), and low-level robot management (RobotLO). This paper details the architecture and functionalities of these levels, emphasizing their roles in optimizing robot behavior on the field through strategic positioning, efficient routing, and precise control mechanisms. Additionally, the report discusses the implementation of the "Bang-bang" trajectory planning for robot movement between two points.**

***Index Terms*—RoboCup Brazil, Wireless Control, Another keyword, RoboCup SSL, Robot Control System, Hierarchical Control, Wireless Communication, Bang-bang Trajectory, Robotics Competition.**

I. Overview of the management system

The control system implements the idea of a hierarchical approach to controlling robots and their interaction in a gaming environment. This idea is based on the division of management tasks into several levels, each of which is responsible for performing local tasks, taking into account the expected performance of the levels below. As part of our study, four key levels were identified: Strategy, Router, high-level (RobotHI) and low-level robot management (RobotLO) (Fig. 1).

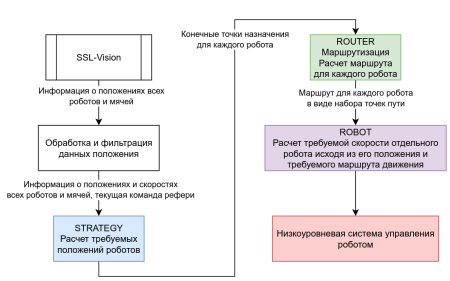

Fig. 1. Control system architecture

A. General description of the levels

1) Strategy: At the strategy level, the tasks of determining the required position of the robot in accordance with the current situation on the field are solved. This level includes modules for predicting the movements of the opponent's robots, analyzing the flight paths of the ball, taking into account risks and forming commands for transfer to the next level (Fig. 2).

2) Router: The routing layer is responsible for building the best path for each robot on the field. The position, speed and obstacles are evaluated, waypoints are formed, which are transmitted to the high-level control of the corresponding robot (Fig. 3).

3) RobotHI: The high-level robot control receives the required number of points and determines the required robot speeds in the global coordinate system and other signals (such as the speed of rotation of the dribbler and commands to kick the ball), which are transmitted to the low-level control system (Fig. 4).

4) RobotLO: The low-level control system, in turn, recalculates the required robot speeds from the global coordinate system to the local SC of the robot and determines the control actions on each of the undercarriage motors.

B. RobotLO level

The RobotLO level receives the required input: the vector of the translational velocity of the robot in the global coordinate system and the angular velocity. The velocity vector is recalculated into the local coordinate system through rotation by the current angle of rotation of the robot ϕ (Fig. 5). After that, the required speeds of each of the motors are calculated according to the kinematics of the chassis and are issued as tasks for engine drivers.

C. RobotHI level

The RobotHI entry level receives the required route of the robot in the form of a set of waypoints. The direction to the nearest waypoint is selected as the direction of the required velocity vector of the robot.

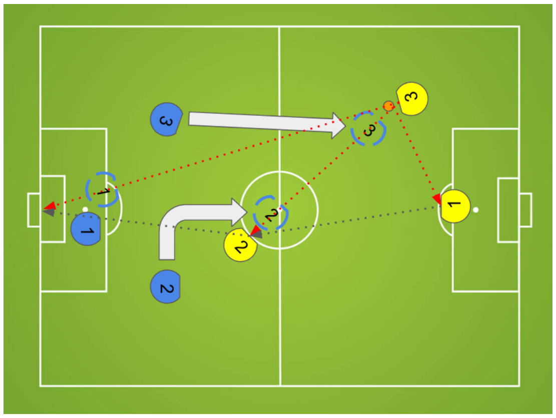

Fig. 2. An illustration of the work of the Strategy level - determining the required positions of allied robots on the field

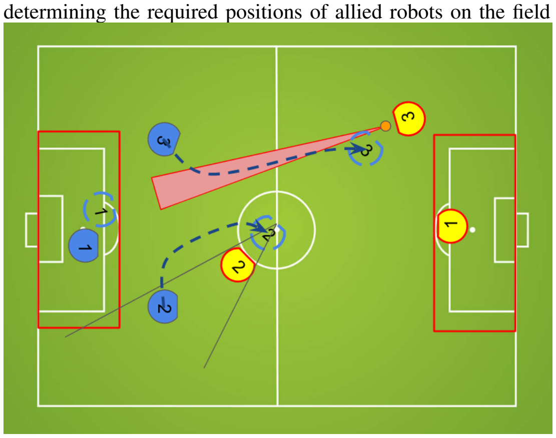

Fig. 3. Illustration of the Router level operation - calculation of optimal routes for each robot in the field

The desired speed of the robot$v_0(t)$ is defined as a function of distance $x(t)$ to the target point using a discretized PID controller with variable structure, saturation and saturation protection by conditional integration. The signal for calculating control actions on a low-level control system in continuous time has the form:

$$v_0(t) = \begin{cases} sat_1[W_1(p)x(t)], & \text{if } x(t) \geq 1.5\text{m} \\ sat_2[W_2(p)x(t)], & \text{if } x(t) < 1.5\text{m} \end{cases},$$

where:

$$W_j(p) = K_j(1 + K_{ij}\frac{1}{p} + K_{dj}\frac{p}{T_f p + 1}),\ j = \{1, 2\},\ p = \frac{d}{dt}$$

$$sat_1[x] = \begin{cases} -1.5\text{m}, & \text{if } x < -1.5\text{m} \\ 1.5\text{m}, & \text{if } x > 1.5\text{m} \\ x, & \text{otherwise} \end{cases},$$

$$sat_2[x] = \begin{cases} -1\text{m}, & \text{if } x < -1\text{m} \\ 1\text{m}, & \text{if } x > 1\text{m} \\ x, & \text{otherwise} \end{cases},$$

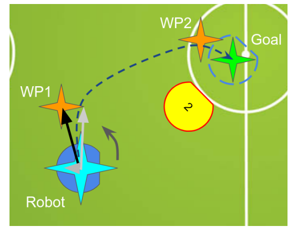

Fig. 4. Illustration of the RobotHI level operation - calculation of the required robot speeds in the global coordinate system based on the current route

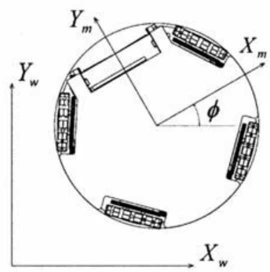

Fig. 5. Global and local coordinate systems of the robot

$sat_1[.]$ is the saturation function at the level of 1.5 m/s, $sat_2[.]$ is the saturation function at the level of 1.0 m/s. The parameters of the regulator are selected experimentally and at the time of writing the report are:

$$\begin{aligned} &K_1 = 6, \quad K_{i1} = 0, \quad K_{d1} = 0.8; \\ &K_2 = 8, \quad K_{i2} = 0, \quad K_{d2} = 0.5. \end{aligned}$$

The quantization period of the system is $T = 0.05$ s. The desired angular velocity of the robot body is calculated using a similar formula with a single set of coefficients:

$$K_1 = 4, \quad K_{i1} = 0.1, \quad K_{d1} = 0.1;$$

and saturation limits $sat_1[.]$ - 30 glad/s. The real distance to the target x(t) and the real speed of the robot v(t) are calculated from signals from video cameras. The

D. Router level

The routing algorithm is responsible for calculating the best route for each robot, taking into account the current situation on the field. There are a large number of algorithms for this purpose, among the most interesting at the moment are the algorithms RRT* and DVG+A*. At the same time, we are using a variation of the pathfinding algorithm using vector fields (Vector Field Pathfinding).

Our implementation of the algorithm allows us to find the shortest path to the end point in a large number of situations, taking into account obstacles on the field. The opponent's robots and gate zones are considered obstacles (for all robots except goalkeepers).

II. Bang-bang Trajectory

A *Bang-bang* trajectory is a path where the acceleration vector has only two possible values.

A. Long Trajectories, the "Prism" Case

The name is derived from the shape of the graph showing the speed projections on the coordinate axes over time.

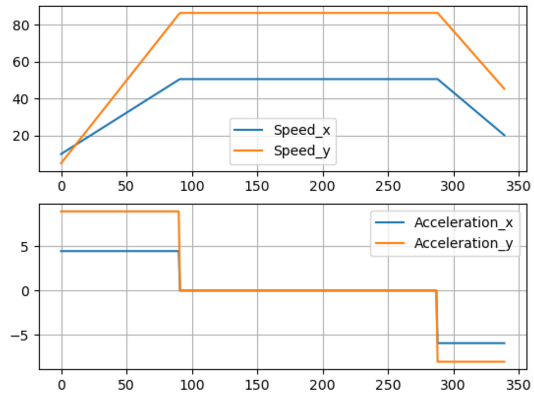

Fig. 6. Graphs with projection

As we can see, both speed projections change simultaneously.

1) Problem Statement: The robot accelerates from an initial speed $\vec{v_1}$ to a maximum speed v with an acceleration $\vec{a_1}$ over time t_1, then travels at maximum speed $\vec{v}$ for time t, and decelerates with an acceleration $\vec{a_2}$ over time t_2 to the final speed $\vec{v_2}$. The robot moves a distance of $\Delta\vec{r}$ during the entire motion.

- Given: a_m (maximum acceleration), v_m (maximum speed), $\vec{v_1}$, $\vec{v_2}$, $\Delta\vec{r}$
- Find: $\vec{a_1}$, $\vec{a_2}$, $\vec{v}$ (recover knowing the magnitudes), t

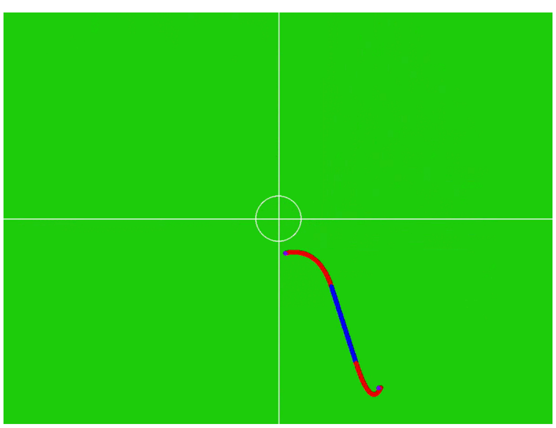

Fig. 7. Choosing an angle

2) Implementation: Red sections indicate acceleration; blue section indicates constant speed; purple points indicate start and end.

We will search by "rotating" the vector v. The other unknowns will be recovered from the formulas (similarly for $t_2, \vec{a_2}$):

$$\vec{a_1} = (\vec{v} - \vec{v_1}) * t_1$$
$$t_1 = \frac{a_m}{|\vec{v} - \vec{v_1}|}$$
$$\vec{a_1} = a_m * \frac{\vec{v} - \vec{v_1}}{|\vec{v} - \vec{v_1}|}$$
$$t = \frac{|\Delta\vec{r} - (\vec{v_1} + \vec{v}) * t_1/2 - (\vec{v_2} + \vec{v}) * t_2/2|}{v_m}$$

To get t, we divide the length of the straight section (the entire movement $\Delta\vec{r}$ minus the movement during acceleration/deceleration) by the speed at which it is passed.

We find the direction of vector v using *ternary search*. To do this, we build the trajectory using the intermediate value of v, and calculate the distance from the end of the resulting trajectory to the required end point.

B. Analyzing the Solution

If the trajectory is found, we finish the algorithm. However, if the trajectory is not found, we return to two cases:

- A trajectory without reaching maximum speed exists, but we couldn't find it.
- Such a trajectory cannot be constructed, so the algorithm with maximum speed should have constructed the route.

In both cases, we need to run both algorithms again, but considering more initial values to accurately find the value around the minimum, which will clearly be less than the others.

C. Short Trajectories, the "Triangle" Case

The name is derived from the shape of the graph showing the speed projections on the coordinate axes over time.

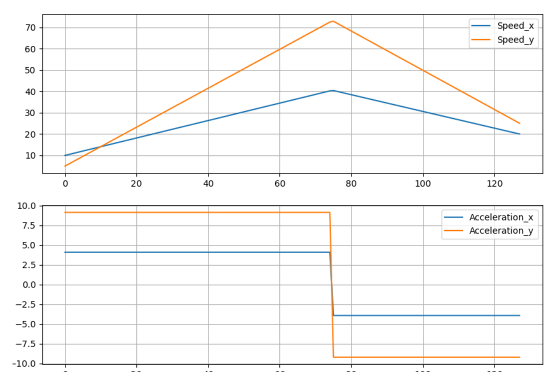

Fig. 8. Graphs with projection

The difference from long trajectories is that the speed reaches its maximum value in the middle of the movement.

1) Problem Statement: The robot accelerates from an initial speed $\vec{v_1}$ with an acceleration $\vec{a_1}$ to a maximum speed v, then decelerates with an acceleration $\vec{a_2}$ to the final speed $\vec{v_2}$. The robot moves a distance of $\Delta\vec{r}$ during the entire motion.

- Given: a_m (maximum acceleration), $\vec{v_1}$, $\vec{v_2}$, $\Delta\vec{r}$
- Find: $\vec{a_1}$, $\vec{a_2}$ (recover knowing the magnitudes), and $\vec{v}$

The task is more complicated than the previous case because we do not know the magnitude of the intermediate speed.

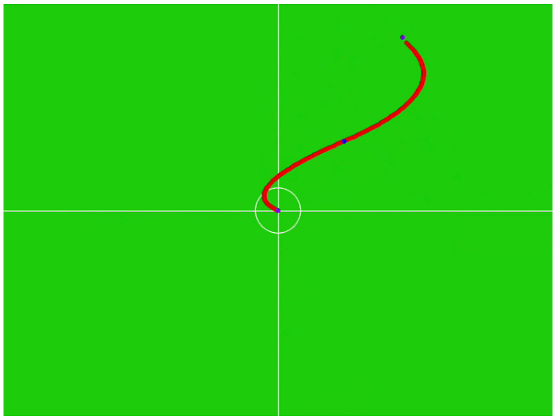

Fig. 9. Choosing an angle

2) Implementation: We iterate the angle for a given speed magnitude. Red sections indicate acceleration; purple points indicate start and end.

We will search by not only "rotating" the vector v, but also iterating the magnitude of its speed. Accelerations and time are recovered from the previously described formulas (see section 1.2).

We will iterate the magnitude of $\vec{v}$ and pass it to the same algorithm that calculated the distance to the target for the "trapezoid".

To find the magnitude, we also iterate 10 values in the interval $[0, v_m]$ and perform a search around the smallest value.

3) Analyzing the Solution: If the trajectory is found, we finish the algorithm. However, if the trajectory is not found, we return to two cases:

- A trajectory without reaching maximum speed exists, but we couldn't find it.
- Such a trajectory cannot be constructed, so the algorithm with maximum speed should have constructed the route.

In both cases, we need to run both algorithms again, but considering more initial values to accurately find the value around the minimum, which will clearly be less than the others.

III. Planning

Unlike last year, where the only thing we could do with the ball was to hit the goal with it (and not really choosing where exactly), we developed several algorithms for both choosing the point for hitting the goal and for passes.

A. Goal kick

We use shadow heuristics to select the point of impact on goal. It is well described in Robocin 2024 Extended TDP[4], our algorithm is very similar.

B. Strategy module

We have also completely rewritten the algorithm for robot control The strategy module can be divided into several parts:

1) Role managment: The algorithm first creates a list of roles to be distributed to the robots, depending on the situation on the field. After that it organises them by priority and assigns each robot (distribution starts with higher priority roles)

2) Creation of waypoints (targets): For each role, the points to which the robots should travel are calculated. After these points are distributed among the robots of this role (if there are several of them).

3) Routing: For each robot the route to its goal is calculated, after that the velocities at the given time are sent to the robot

IV. Conclusion

In conclusion, the SPbUnited team's robot control system demonstrates a robust and effective approach to managing autonomous robots in a competitive environment. By implementing a hierarchical control architecture, the system ensures optimal coordination and performance across different levels of control. The integration of advanced strategies, efficient routing algorithms, and precise low-level control mechanisms allows for responsive and adaptive robot behavior on the field. The utilization of "Bang-bang" trajectory planning further enhances the robots' ability to execute complex maneuvers efficiently. As a result, the SPbUnited team is well-equipped to compete in the RoboCup SSL, showcasing innovations that contribute to the advancement of robotics technology in competitive settings. Future work will focus on refining these systems and exploring new methodologies to continually improve the team's performance.

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References

[1] Nicolai Ommer, Andre Ryll, Mark Geiger, TIGERs Mannheim 2019 Extended Team Description Paper

[2] Andreas Wendler, Tobias Heineken, ER-Force 2020 Extended Team Description Paper

[3] Alexandr Fradkov, Sergey Filippov, Alexandr Meshcheryakov, Boris Viktorov, Iulia Merzlyakova, Ilya Ustinov, Vasilii Ivanov, Mikhail Lipkovich, and Yurii Glazov, SPbUnited 2023 Team Description Paper

[4] Alberto Franca, Beatriz Barros, Cauˆe Gomes, Cec´ılia Silva, Charles M. Alves,Davi C. Barbosa, Driele Xavier, Elisson Ara´ujo, Felipe Pereira, Hierlan A.Batista, Lucas H. Cavalcanti, Marcela Asfora, Matheus Alves, Matheus Paix˜ao,Matheus Vasconcelos, Matheus Vin´ıcius, Jo˜ao G. Melo, Jos´e R. Silva, Jos´e V.Cruz, J´ulia Leite, Pedro H. Santana, Pedro P. Oliveira, Riei Rodrigues, RyanMorais, Tamara Teobaldo, Vin´ıcius Dutra, Victor Ara´ujo, and Edna Barros, RobˆoCIn 2024 Extended Team Description Paper

[5] TIGERs Mannheim github: https://github.com/TIGERs-Mannheim

[6] ZJUNlict github: https://github.com/ZJUNlict